

Rainul Hakim

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EDUCATION

Davidson College

Bachelor of Science, Computer Science / GPA: 3.76

- Relevant Coursework: Data Structures, Programming and Problem Solving, Database Programming, Linear Algebra, Calculus I & Modeling, Calculus II

Davidson, NC

Expected May 2029

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

ML / Scientific Computing: PyTorch, PyTorch Geometric, TensorFlow.js, MediaPipe, NumPy, scikit-learn

Web: React, Next.js, Node.js, Supabase, PostgreSQL, Vercel Serverless Functions, HTML, CSS

Tools: Git, GitHub, VS Code, HPC clusters

EXPERIENCE

Machine Learning Research Assistant

May 2026 – Aug 2026

Facility for Rare Isotope Beams (FRIB)

East Lansing, MI & Davidson, NC

- Pretrained a DGCNN encoder with SimCLR-style contrastive self-supervised learning on **60,795** unlabeled AT-TPC point-cloud events, learning representations of collision tracks without manual labels, toward a foundation model adaptable to multiple downstream physics tasks
- Fine-tuned and evaluated the pretrained encoder via linear probing on a 2,458-event labeled subset for 3-class track-multiplicity classification, reaching **88.82%** accuracy and **88.60%** weighted F1
- Ran systematic augmentation ablations for the pretraining stage and designed a physically-motivated rotation scheme that outperformed generic augmentations
- Trained on real experimental data with **PyTorch** and **PyTorch Geometric** on an HPC cluster; contributed reproducible, documented code to a shared cross-institutional codebase

Student Consultant

Aug 2026 – Present

Data CATS (Consulting, Analytics, Tutoring Services), Davidson College

- Supported **85+** students and faculty on data analytics, from framing questions to collecting, visualizing, and interpreting data

NEXT Level Student Consultant

Feb 2026 – May 2026

The Jay Hurt Hub for Innovation and Entrepreneurship

- Built the **Next.js/React** front end for an AI study planner, integrating the **API** via TypeScript API routes
- Designed prompts, **Zod**-validated output schemas, and wellbeing guardrails; QA-tested via demo scenarios and client feedback
- Delivered a working prototype to a startup client, drawing follow-up interest from Davidson faculty and other consultant teams

PROJECTS

Campus Event Discovery Platform | *Python, PostgreSQL, AWS Lambda, Amazon RDS, API Gateway, Mapbox*

- Built event discovery platform ingesting Davidson event data into a relational **PostgreSQL** database
- Automated AWS pipeline with **EventBridge** and **Lambda** to extract, clean, and update event and location data
- Developed **React/Mapbox** interface to filter events by location, date, and category with campus navigation

Left No Crumbs! – 1st Place (\$1,000), IdeaSprint | *Next.js, Supabase, PostgreSQL, Web Push*

- Co-founded and built a real-time food-rescue platform alerting **600+** students across **55** organizations to surplus catered meals
- Designed a **PostgreSQL** schema with row-level security, campus-only auth, dietary filtering, and rate-limited push notifications; partnered with Davidson Dining and T&I on Davidson One integration

AI Meal Analysis & Recommendation Platform | *Python, Gemini API, React, Vite, JavaScript, Vercel*

- Built AI meal analysis pipeline generating macro breakdowns, health flags, and personalized alternatives
- Integrated **Gemini API** behind serverless functions and implemented meal comparison and weekly scoring systems
- Deployed React application used by **100+** Davidson students to evaluate and compare campus meals

In-Browser Hand Gesture Recognition (ML) | *JavaScript, TensorFlow.js, MediaPipe*

Open-source extension

- Built a gesture-triggered effects app running fully in-browser at ~30 fps with no backend: MediaPipe Holistic extracts 126 hand-landmark coordinates per frame and feeds a TensorFlow.js neural network
- Normalized landmarks by wrist position and scale for hand-size invariance; trained a binary classifier (Dropout 0.3, 50 epochs, 0.97 confidence threshold) with a custom in-browser training UI
- Segmented the user's silhouette with MediaPipe Selfie Segmentation and rendered multiple clones with smoke effects on Canvas

LEADERSHIP & INVOLVEMENT

Resident Advisor, Davidson College

2026 – Present

Logistics, Hack@Davidson

2026 – Present

Event Coordinator, Davidson Entrepreneurship Club

2025 – Present

Secretary, Association for Computing Machinery (ACM), Davidson College

2026 – Present

Treasurer, Davidson International Association

2026 – Present